

Proposal for Upgrading Firewall for Fortigate Series

Project Description

This Firewall firmware 5.0.9 was old present moment, but we can firmware upgrade to 6.2.7, meaning that we have use to address newly-provisions security and feature vulnerabilities over the new firmware. Additionally, besides possessing all the right features the new clear and more user-friendly interface of the apps is designed to assist you and help you work more efficiently than ever.

Project Plan

First step: ready for update

Available before Upgrade, we are always available for Upgrade Firewall

- Laptop window, putty software or security CRT
- Console cable USB to C
- Cable RJ45
- Account download the firmware file from the Fortinet Customer Service & Support and Double check Upgrade Patch

Verify Patch before upgrade, it's patch update recommend follow table as bellow:

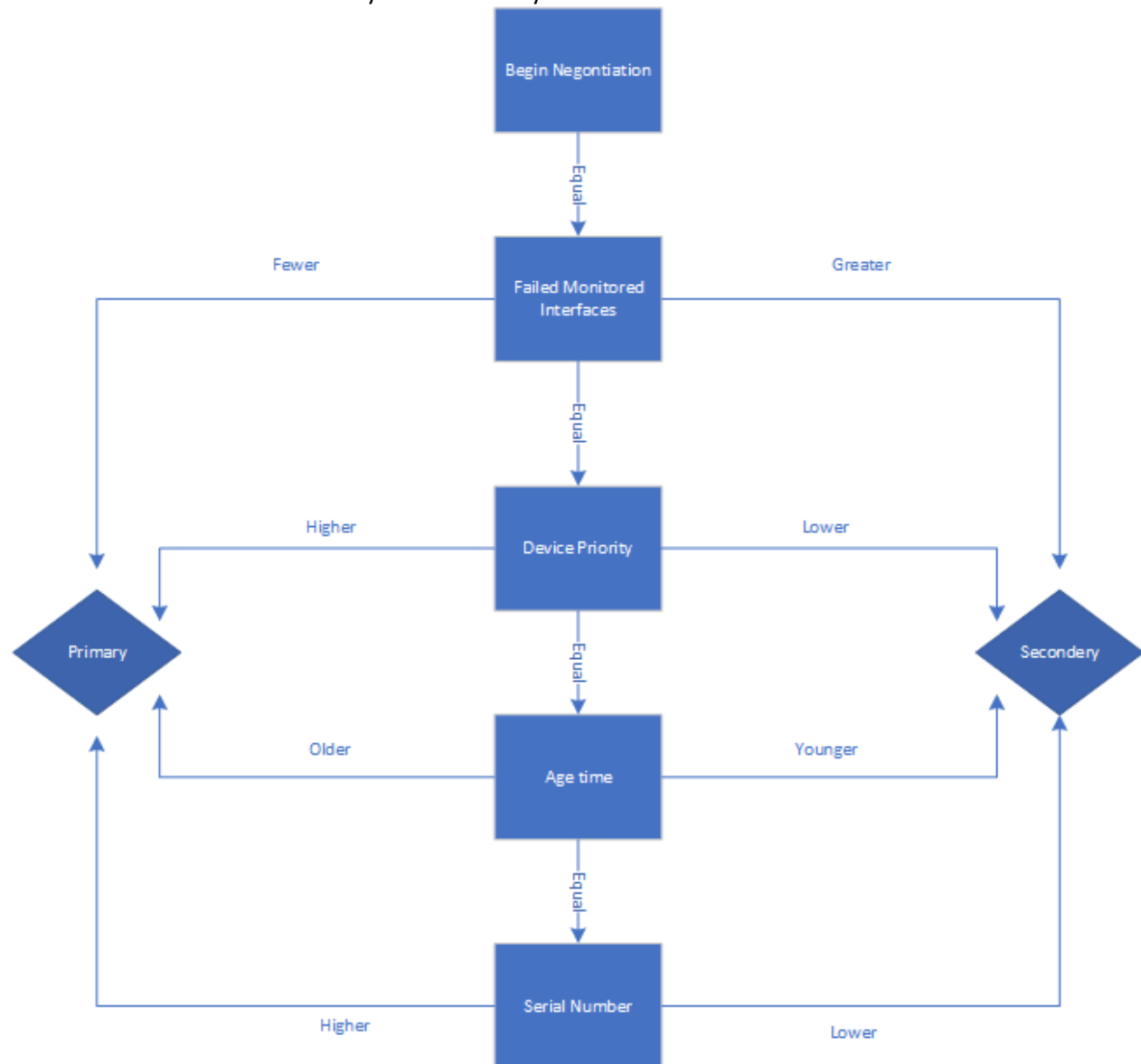


Patch table information trusted, if firmware as 5.0.9

Patch	Build
5.0.9	292
5.0.11	310
5.2.5 or 5.2.6	701 or 711
5.4.0	1011
5.4.2	1100
5.4.4	1117
5.6.2	1486
5.6.6	1630
5.6.8	1672

5.6.11	1700
6.0.10	0365
6.2.7	1190

Behavior HA cluster vote Primary and Secondary



When HA configured, the Firewall in one cluster will vote Primary and secondary. If no configuration monitor interfaces or all interfaces monitored status good, next step compare Device Priority, look image

information above, Higher priority will become Primary Unit and reverse. If parameters is Equal, the cluster considers uptime first (Age) and final is Serial Number.

Step 2: Check Configuration and Backup

First, login Fortigate using Mgmt Port or Console port, next step check HA Override status:

- *If Override is enabled, the cluster considers the Device Priority setting. Both nodes usually make a second failover in order to resume their original roles.*
- *If Override is disabled, the cluster considers uptime first. The original primary node will have a smaller uptime due to the order of reboots during the firmware upgrade. Therefore it will not resume its active role; instead, the node with the greatest uptime will remain the new primary node. A second failover will not occur.*

Check Diagram Network for High Availability ready for lost connection from Primary Unit or restart Primary, make sure the secondary will be ready to take on the primary.

Configuration backup the Primary device's, ensure file back up can read and write using notepad++ for tshoot if must recover configuration or re-config.

** Note: mode Active - Passive or mode Active - Active same process update*

Step 3: Upgrade Process

Method one, update firmware to Primary Unit will be auto deployment to Secondary Fw

Load firmware into Fortigate using web-client from ip address port Mgmt, fortigate will be auto check the firmware before updating, processing will be start when firmware accepted:

1. *The primary node pushes the firmware image to the member nodes.*
2. *The primary node notifies the member nodes of the upgrade, and it takes their user traffic during the upgrade.*
3. *The upgrade command is run on the member nodes, the systems are rebooted, and the member nodes send the primary node an*
4. *The upgrade command is run on the primary node, and it reboots. When the system is rebooting, a member node assumes primary status, and the traffic fails over from the former primary node to the new primary node.*
5. *After every image version upgrade kindly make sure both devices are in sync before proceeding for further upgrades.*

Reboot times vary by the appliance model, and also by differences between the original firmware version and the firmware version you are installing. However, it will take around 30 minutes to upgrade the HA devices and get in sync.

Note: Do not apply process above for downgrade Firmware because:

If downgrading to a previous version, do not use this procedure. The HA node unit on a member node might detect that the primary node has older firmware, and attempt to upgrade it to bring it into sync, undoing your downgrade. Instead, switch out of HA, downgrade each node individually, then switch them back into HA mode.

Method two, upgrade each node individually or downgrade

1. *Equipment must be labelled*
2. *Disconnected all Ports, switch Secondary node out of HA*
3. *Load Firmware into Secondary node follow patch update, fortigate will be auto check the firmware before updating, processing will be start when firmware accepted*
4. *Fortigate will be stop working when Firmware archive completed and reboot immediate*
5. *Check again configuration original, and continue implementation procession update.*

When finish updated firmware on the Fortigate Secondary unit, start up unit and rollback connection ports to original. But, except port HA and shutdown the Primary unit working. Secondary Unit will be become the Primary and it takes their user traffic during the upgrade Primary original.

Continue to update as above steps, as mentioned.

Troubleshoot

when firmware update won't lose configuration, but if the configuration is lost, we will restore backup file or re-config the device based on the backup file.

If the software is not accepted, we will check the Upgrade patch file or use another authorized firmware accepted.

We will not be responsible for hardware failure.